THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE–REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

THE CITY COUNCIL OF DODOMA (CCD)

FORM ONE TERMINAL EXAMINATION 2022

032-CHEMISTRY

TIME:2:30Hours **JULY,2022**

INSTRUTIONS

1. This paper consists of section A and B with a total of ten (10) questions.
2. Answer all question in section A and B.
3. Cellular phones, calculators and any unauthorized materials are not allowed in the examination room.
4. Write your examination number on every page of your answer sheet(s).

SECTION A

Answer all questions in this section

1. For each of the items (i –x), choose the correct answer from among the given alternatives and write its letter in the box provided.

(i) When a burning fuel produces blue colour it means there is
(a) Adequate supply of oxygen with production of soot.
(b) Inadequate supply of oxygen without production of soot.
(c) Adequate supply of oxygen with production of less heat.
(d) Adequate supply of oxygen with production of more heat.

(ii) Bertha saw the "flammable sign" on a box. She decided the following possible interpretation, which among the following is the most correct one?

(a) The box contain papers
(b) The box contain juice
(c) The box contain radioactive materials
(d) The box contain petrol

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(iii) A student hypothesized that the amount of sunlight a sunflower plant receives determines the number of sunflower seeds the plant produces. In her experiment the number of seeds is the…..

(a) Independent variable
(b) Dependent variable
(c) Controlled variable

(d) Uncontrolled variable.

(iv) Which among the following substances usually burn without producing flame?

(a) Candle and coal
(b) Fire wood and charcoal
(c) Paper and matchstick
(d) Coal and charcoal

(v) A mixture of water and alcohol was carefully distilled. Which property of the first drops of the liquid distillate collected is **not** correct? The first drops
(a) Have the same boiling point as water
(b) Mix easily with water
(c) Are colour less
(d) Will boil at a temperature less than the boiling point of water.

(vi) When you melt a piece of iron it undergoes;

(a) Sublimation
(b) Chemical change
(c) Physical change
(d) Combination

(vii) Which of the following warning signs is likely to appear on the bottle containing concentrated nitric acid in the laboratory?

(a) Corrosive
(b) Flammable
(c) Oxidant
(d) Toxic

(viii) In the scientific study the tentative explanation for the observed chemical phenomenon can be proved by;

(a) Data analysis
(b) Hypothesis
(c) Experimentation
(d) Conclusion

(ix) In the Bunsen burner a sooty flame is most likely to be formed when the;

(a) Air holes are fully closed
(b) Flame is noisy.
(c) Air holes are fully opened
(d) Flame is bigger and hotter

(x) The following is a set of apparatus which are used for heating.

(a) Evaporation dish, tongs and retort stand

(b) Test tube, pipette and Bunsen burner

(c) Crucible, evaporation dish and boiling tube.

(d) Safety goggles, evaporation dish and dropper.

ANSWERS.

I	Ii	iii	iv	v	Vi	vii	viii	ix	x

2. (a) Match the physical processes represented by arrows (i) - (v) in **List A** with the corresponding items in **List B** by writing the letter of the correct response beside the item number in the answer booklet provided.

List A	List B
i. A substance that has no definite shape and size ii. An apparatus used for keeping substances dry. iii. Used for reducing muscle pain iv. Contains guidelines on how to use the item in the first aid. v. Blockage of the upper part of the airway by food or other object.	A. Safety pin B. Solid C. Plaster. D. First aid manual E. Liniment. F. Iodine tincture G. Desiccator. H. Test tube rack. I. Gas J. Suffocation K. Choking.

ANSWERS.

LIST A	I	Ii	Iii	iv	V
LIST B					

(b) Fill in the blank spaces by using the appropriate terms.

(i) The factors that affect the problem being investigated are referred to as…….

(ii) The only known liquid metal is ……

(iii) A flame is a region of burning gases that produces ……and ……..

(iv) Burns caused by hot liquids or vapours are called……

(v) Used for holding hot apparatus and substances…….

SECTION B

Answer all questions

3. (a) Chemistry is one of the important subjects in our daily life activities. Outline its importance using the following sub headings as guidelines.

Introduction.

……
……
……

Medicine.

……
……
……

Agriculture.

……
……
……

(b) Complete the table below on the different products made by application of chemistry knowledge.

Field	Products
Home care and cosmetics industry	i. ii.
Food and beverage industry	i. ii.

4. A form one student is asked to put labels on reagent bottles in the laboratory; the labels should contain warning sign names and symbols. Assume you are the one asked, fill the table below.

S/NO	The bottles containing	Warning sign name	The chemical symbol
1.	Spirit		
2.	Concentrated sulphuric acid		
3.	Hydrogen peroxide		

4.	Mercury		

5. (a) Outline three apparatus that are made up of clay.

(i) ……..
(ii) ……
(iii)……..

(b) Imagine you want to perform an experiment in the laboratory, name the apparatus to be used for the following works.

(i) Grinding solid substances into fine powder…….
(ii) Storing distilled water……..
(iii) Measuring the temperature of the liquid…….

(c) Briefly explain the following terms.

(i) First aid

……
……

(ii) First aid kit.

……
……

6. Nusulupilya wants to investigate on whether temperature affects the solubility of sugar in water. As the only chemistry expert available, briefly advise him by giving procedures on how he can systematic conduct his investigation.

7. (a) Assume that you are doing an experiment in the laboratory at 08:30Pm and suddenly the light goes off. Give two reasons to justify the fact that you would consider luminous flame rather than non-luminous flame as an alternative source for lighting.

i. ……...ii……

b) Identify two properties of the flame produced by the Bunsen burner (air holes fully opened) that could not be found in the flame produced by the spirit burner.

i. ……ii……..

c) Draw a spirit lamp flame and label the different zone

8. (a) Briefly explain the following terms.

(i) Sublimation

(ii) Deposition.

(b) Mention three substances that can undergo the change in 7 (a) (i).

(c) The change of state of matter has great significance in our daily life. Explain this statement basing on;

(i) Water cycle

(ii) Drying of materials

(iii) Metallurgy

9. (a)In how ways can you help a victim experiencing fainting. Outline four steps.

(b) Striking back of a Bunsen burner happens when a flame passed down the barrel and burns at the air holes. Referring to this fact write the cause(s) and how to control it.

10. (a) Briefly explain the following concepts.

(i) Element

(iii) Compound

(iv) Mixture.

(v) Solution.

(b) A teaspoon of salt is added in water in beaker A and a teaspoonful of chalk powder is added in water in beaker B. Give three (3) differences between the mixture in beaker A and beaker B.

Breaker A	Beaker B